Handout 9: SAN and UCI cheat sheet

People write moves in **standard algebraic notation**, SAN. Chess programs talk to each other in **UCI** notation. Remus reads and writes both at its edges and uses `(row, col)` pairs inside. `learn_chess.md` explains every line of this table with examples.

Piece	Letter in SAN	Letter on the Remus board
King	`K`	`K` White, `k` Black
Queen	`Q`	`Q` White, `q` Black
Rook	`R`	`R` White, `r` Black
Bishop	`B`	`B` White, `b` Black
Knight	`N`	`N` White, `n` Black
Pawn	no letter	`P` White, `p` Black
Empty square		`.`

What happened	SAN	UCI	Remus inside
Pawn from e2 to e4	`e4`	`e2e4`	`Move((6, 4), (4, 4))`
Knight from g1 to f3	`Nf3`	`g1f3`	`Move((7, 6), (5, 5))`
Bishop on c4 takes the piece on f7	`Bxf7`	`c4f7`	`Move((4, 2), (1, 5))`
Pawn on e5 takes on d6 (also en passant)	`exd6`	`e5d6`	`Move((3, 4), (2, 3))`
Two knights could reach d2; the one from b1 goes	`Nbd2`	`b1d2`	`Move((7, 1), (6, 3))`
Pawn from e7 reaches e8 and becomes a queen	`e8=Q`	`e7e8q`	`Move((1, 4), (0, 4), "Q")`
The same pawn becomes a knight	`e8=N`	`e7e8n`	`Move((1, 4), (0, 4), "N")`
White castles short (king e1 to g1)	`O-O`	`e1g1`	`Move((7, 4), (7, 6))`
White castles long (king e1 to c1)	`O-O-O`	`e1c1`	`Move((7, 4), (7, 2))`
A move that gives check	add `+`, as in `Bb5+`	nothing added	nothing added
A move that gives checkmate	add `#`, as in `Qh7#`	nothing added	nothing added
No legal move to send		`0000`	

The square names and the pairs:

Square	`(row, col)`	Square	`(row, col)`
`a8`	`(0, 0)`	`h8`	`(0, 7)`
`a1`	`(7, 0)`	`h1`	`(7, 7)`
`e4`	`(4, 4)`	`d5`	`(3, 3)`
`e2`	`(6, 4)`	`e7`	`(1, 4)`

`row = 8 - rank` and `col = "abcdefgh".index(file)`. Row 0 is rank 8, because Remus prints the board with Black's side at the top, the way a chess book does.